FORTISAID LAB 1

Azure RBAC Walkthrough: Users + Groups + Group-Based RBAC (Portal, PowerShell, Bash)

What I built

A proof-of-concept showing how to:

1. Create users and groups in **Microsoft Entra ID**
2. Assign users to groups
3. Assign an Azure RBAC role to a **group** (not a user)
4. Verify effective access at a **resource group scope** (`FortisAid-Lab`, East US)

Built to demonstrates **least privilege** and reduces permission sprawl by using group-based access control.

Step 0: Target Architecture (What I intended)

Users

- Emmanuel Tobi (Senior Admin)
- Tobi Babalola (Junior Admin)
- Eneattah Grace (Service Desk)

Groups

- Senior Admins
- Junior Admins
- Service Desk

RBAC

- Assign **Virtual Machine Contributor** → to **Service Desk group**
- Scope: **Resource Group** `FortisAid-Lab`

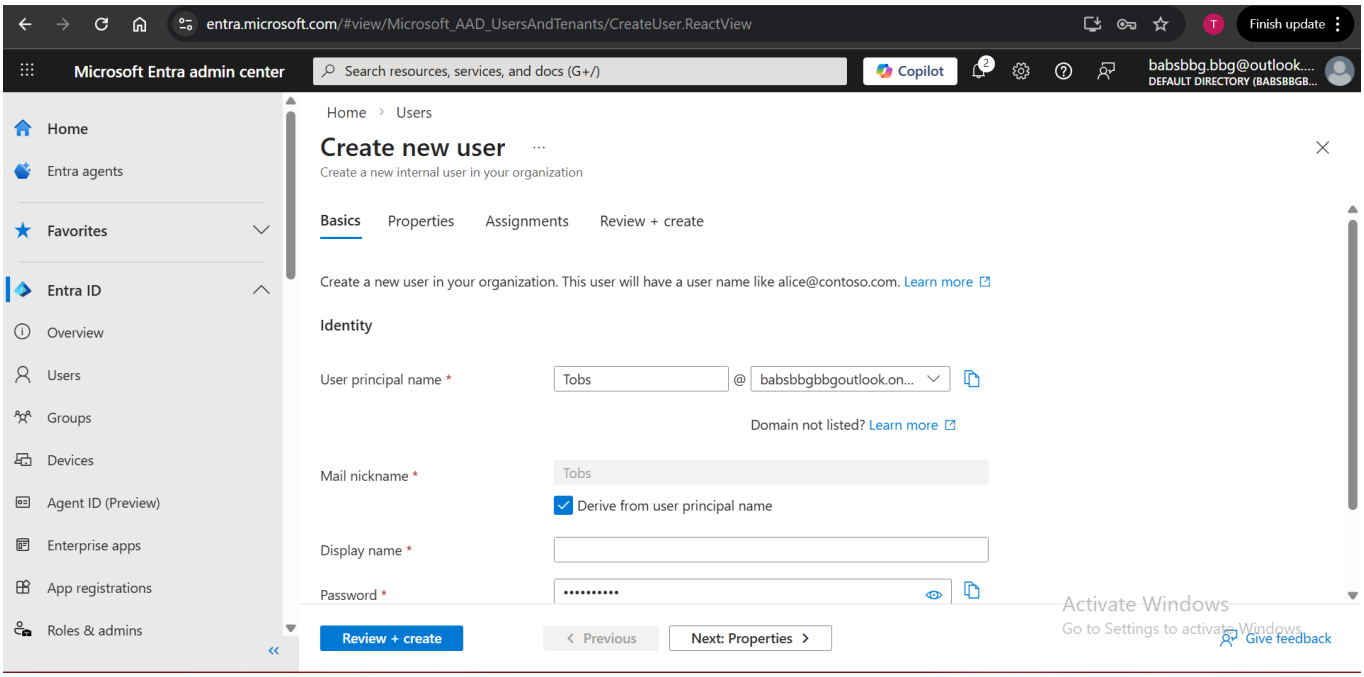
"Pasted image 20260204113056.png" is not created yet. Click to create.

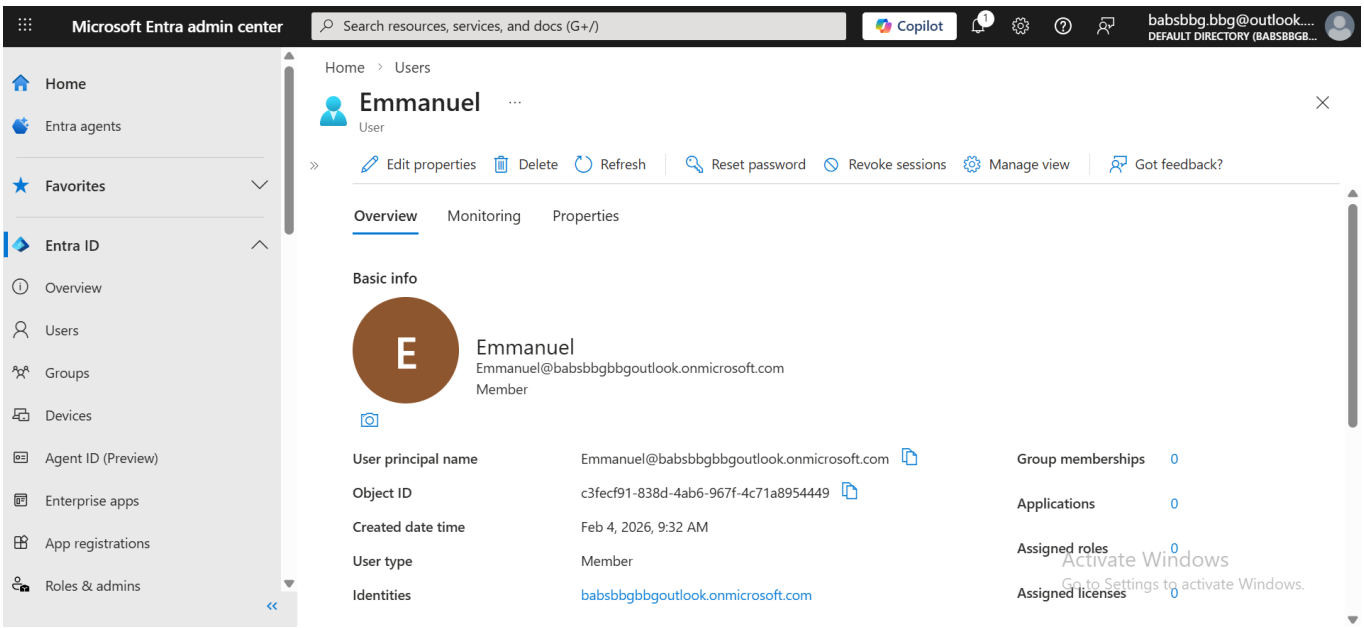
Step 1: Create Senior Admin user (Portal) — Emmanuel Tobi

Steps (Azure Portal → Entra ID)

1. Sign in to `https://portal.azure.com`
2. Search **Microsoft Entra ID**
3. Go to **Manage** → **Users**
4. Click **+ New user** → **Create new user**
5. Fill in:
   - **User name:** `emmanuel.tobi@<tenant>.onmicrosoft.com`
   - **Name:** `Emmanuel Tobi`
   - **Password:** Auto-generate (save it)
   - **Account enabled:** Yes
6. Click **Create**

Verification:
Microsoft Entra admin center → Users → **Emmanuel Tobi** appears.





Step 2: Create Junior Admin user (PowerShell) — Tobi Babalola

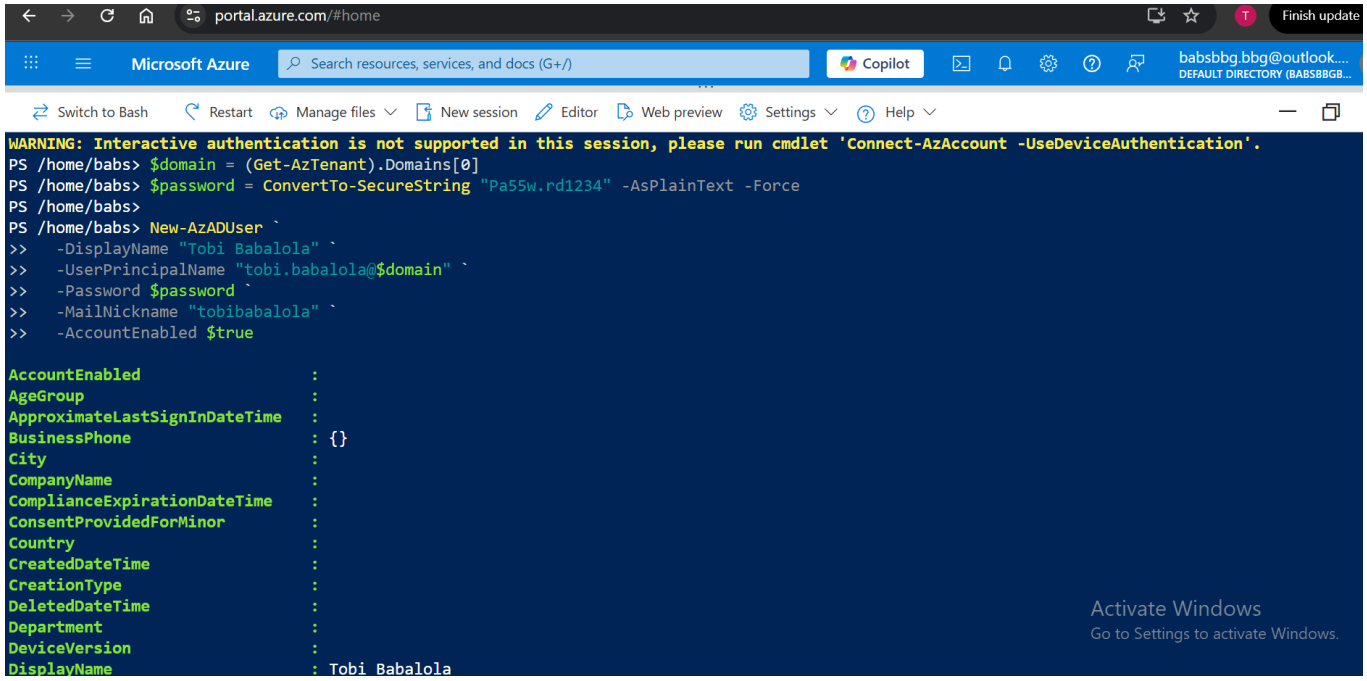
Actions (Cloud Shell → PowerShell)

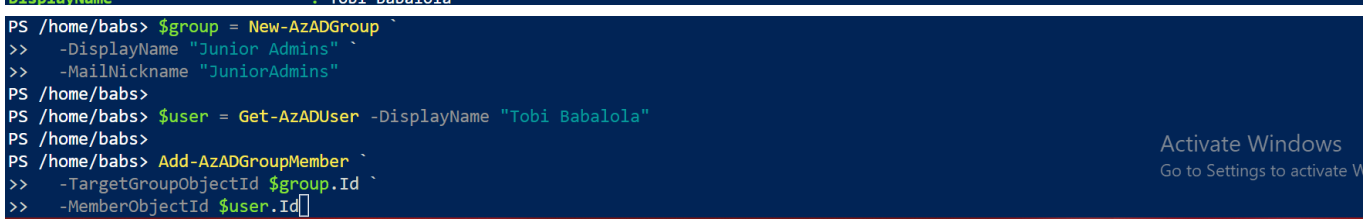
```
Connect-AzAccount
$domain = (Get-AzTenant).Domains[0]
$password = ConvertTo-SecureString "Pa55w.rd1234" -AsPlainText -Force

New-AzADUser `
  -DisplayName "Tobi Babalola" `
  -UserPrincipalName "tobi.babalola@$domain" `
  -Password $password `
```

```
  -MailNickname "tobibabalola" `
  -AccountEnabled $true
```

Verification:

```
Get-AzADUser -DisplayName "Tobi Babalola"
```





Step 3: Create Service Desk user (Bash / Azure CLI) — Eneattah Grace

Important note

I hit a UPN domain error until I used a **verified tenant domain**. The verified domain I used was:

`babsbbgbbgoutlook.onmicrosoft.com`

Actions (Cloud Shell → Bash)

```
DOMAIN="babsbbgbbgoutlook.onmicrosoft.com"

az ad user create \
  --display-name "Eneattah Grace" \
```

```
  --user-principal-name eneattah.grace@$DOMAIN \
  --password "Pa55w.rd1234"
```

Verification:

```
az ad user show --id eneattah.grace@$DOMAIN --query displayName -o tsv
```

Expected output:

`Eneattah Grace`

```
babs [ ~ ]$ DOMAIN="babsbbgbbgoutlook.onmicrosoft.com"

az ad user create \
  --display-name "Eneattah Grace" \
  --user-principal-name eneattah.grace@$DOMAIN \
  --password "Pa55w.rd1234"
{
  "@odata.context": "https://graph.microsoft.com/v1.0/$metadata#users/$entity",
  "businessPhones": [],
  "displayName": "Eneattah Grace",
  "givenName": null,
  "id": "e81194e0-20e8-4b87-bd52-6731a2f2b143",
  "jobTitle": null,
  "mail": null,
  "mobilePhone": null,
  "officeLocation": null,
  "preferredLanguage": null,
  "surname": null,
  "userPrincipalName": "eneattah.grace@babsbbgbbgoutlook.onmicrosoft.com"
}
babs [ ~ ]$ az ad user show --id eneattah.grace@$DOMAIN --query displayName -o tsv
Eneattah Grace
babs [ ~ ]$
```

Step 4: Create groups and add members (Entra ID)

4A) Create Senior Admins group + add Emmanuel Tobi (Portal)

1. Entra ID → **Groups** → **+ New group**
2. Type: **Security**
3. Name: **Senior Admins**
4. Membership type: **Assigned**
5. Add member: **Emmanuel Tobi**
6. Create

Microsoft Entra admin center

Search resources, services, and docs (G+/)

Copilot

babsbbg.bbg@outlook.... DEFAULT DIRECTORY (BABSBBGB...

Home
Entra agents
Favorites
Entra ID
Overview
Users
Groups
Devices
Agent ID (Preview)
Enterprise apps
App registrations
Roles & admins

Home > Groups | Overview > New Group > Groups | All groups

New Group

Got feedback?

Group type *
Security

Group name *
FortisAid

Group description
Enter a description for the group

Membership type
Assigned

Owners
1 owner selected

Members

Create

Activate Windows
Go to Settings to activate Windows.

Microsoft Entra admin center

Search resources, services, and docs (G+/)

Copilot

babsbbg.bbg@outlook.... DEFAULT DIRECTORY (BABSBBGB...

Home
Entra agents
Favorites
Entra ID
Overview
Users
Groups
Devices
Agent ID (Preview)
Enterprise apps
App registrations
Roles & admins

Home > Groups | Overview > New Group > Groups

Groups | All groups

Default Directory

Overview
All groups
Insights (Preview)
Deleted groups
Diagnose and solve problems

Settings
General
Expiration
Naming policy

Activity
Privileged Identity Management
Access reviews
Audit logs

New group | Download groups | Refresh | Manage view | Delete

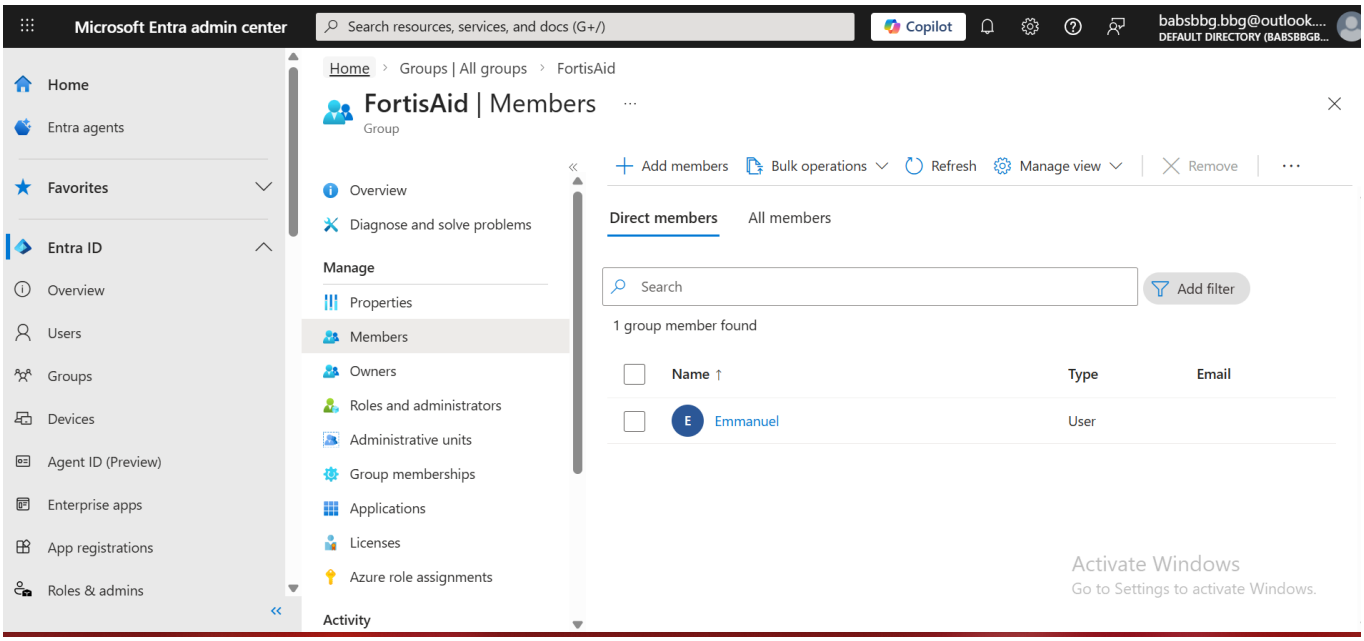
Search | Add filter

Search mode Contains

1 group found

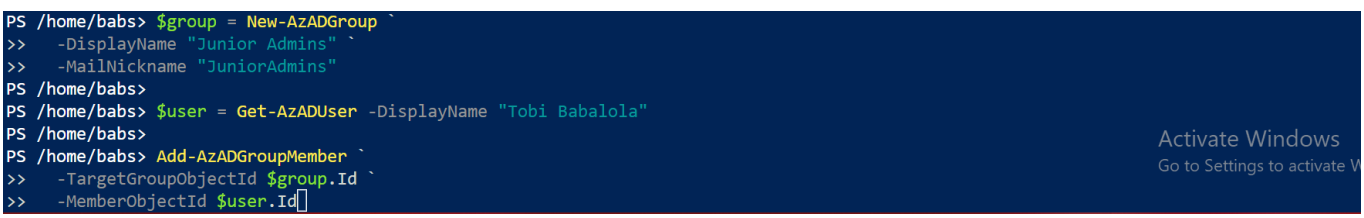
Name	Object Id
FS FortisAid: Senior Admins	7606b0a4-3041-4089-9241-401d640b8339

Activate Windows
Go to Settings to activate Windows.



4B) Create Junior Admins group + add Tobi Babalola (PowerShell)

```
$group = New-AzADGroup `
  -DisplayName "Junior Admins" `
  -MailNickname "JuniorAdmins"

$user = Get-AzADUser -DisplayName "Tobi Babalola"

Add-AzADGroupMember `
  -TargetGroupObjectId $group.Id `
  -MemberObjectId $user.Id
```



4C) Create Service Desk group + add Eneattah Grace (Bash)

```
az ad group create \
  --display-name "Service Desk" \
  --mail-nickname servicedesk

USER_ID=$(az ad user show \
  --id eneattah.grace@babsbbgbbgoutlook.onmicrosoft.com \
  --query id -o tsv)
```

```
GROUP_ID=$(az ad group show \
  --group "Service Desk" \
  --query id -o tsv)

az ad group member add \
  --group $GROUP_ID \
  --member-id $USER_ID
```

Verification:

```
az ad group member list \
  --group "Service Desk" \
  --query "[].displayName" -o table
```

Microsoft Azure | Search resources, services, and docs (G+/) | Copilot

Switch to PowerShell | Restart | Manage files | New session | Editor | Web preview | Settings | Help

```
babs [ ~ ]$ az ad group create \
  --display-name "Service Desk" \
  --mail-nickname servicedesk
{
  "@odata.context": "https://graph.microsoft.com/v1.0/$metadata#groups/$entity",
  "classification": null,
  "createdDateTime": "2026-02-04T09:00:40Z",
  "creationOptions": [],
  "deletedDateTime": null,
  "description": null,
  "displayName": "Service Desk",
  "expirationDateTime": null,
  "groupTypes": [],
  "id": "e46addca-dbf0-4508-b81a-4ab5fe7770a7",
  "isAssignableToRole": null,
  "mail": null,
  "mailEnabled": false,
  "mailNickname": "servicedesk",
  "membershipRule": null,
  "membershipRuleProcessingState": null,
  "onPremisesDomainName": null,
  "onPremisesLastSyncDateTime": null,
  "onPremisesNetBiosName": null,
  "onPremisesProvisioningErrors": [],
  "onPremisesSamAccountName": null,
  "onPremisesSecurityIdentifier": null,
  "onPremisesSyncEnabled": null,
  "preferredDataLocation": null,
  "preferredLanguage": null,
```

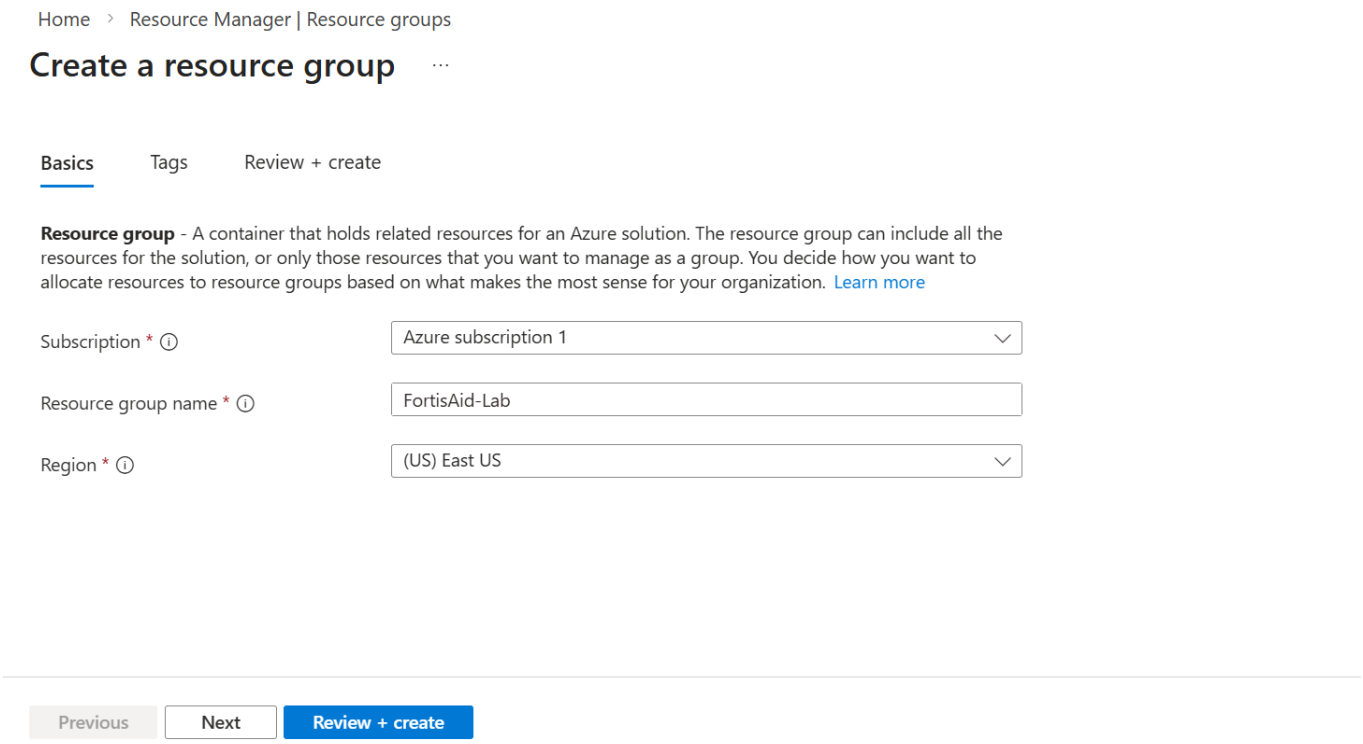
```
babs [ ~ ]$ USER_ID=$(az ad user show \
  --id eneattah.grace@babsbbgbbgoutlook.onmicrosoft.com \
  --query id -o tsv)

GROUP_ID=$(az ad group show \
  --group "Service Desk" \
  --query id -o tsv)
babs [ ~ ]$ az ad group member add \
  --group $GROUP_ID \
  --member-id $USER_ID
babs [ ~ ]$ az ad group member list \
  --group "Service Desk" \
  --query "[].displayName" -o table
Result
--------------
Eneattah Grace
babs [ ~ ]$
```

Step 5: Assign RBAC Role to the Service Desk group (Scope: FortisAid-Lab)

Goal

Service Desk can manage VMs but can't do subscription-wide admin tasks.

1. Azure Portal → Resource groups → **FortisAid-Lab**
2. **Access control (IAM) → + Add → Add role assignment**
3. Role: **Virtual Machine Contributor**
4. Assign access to: **User, group, or service principal**
5. Select: **Service Desk**
6. **Review + assign**



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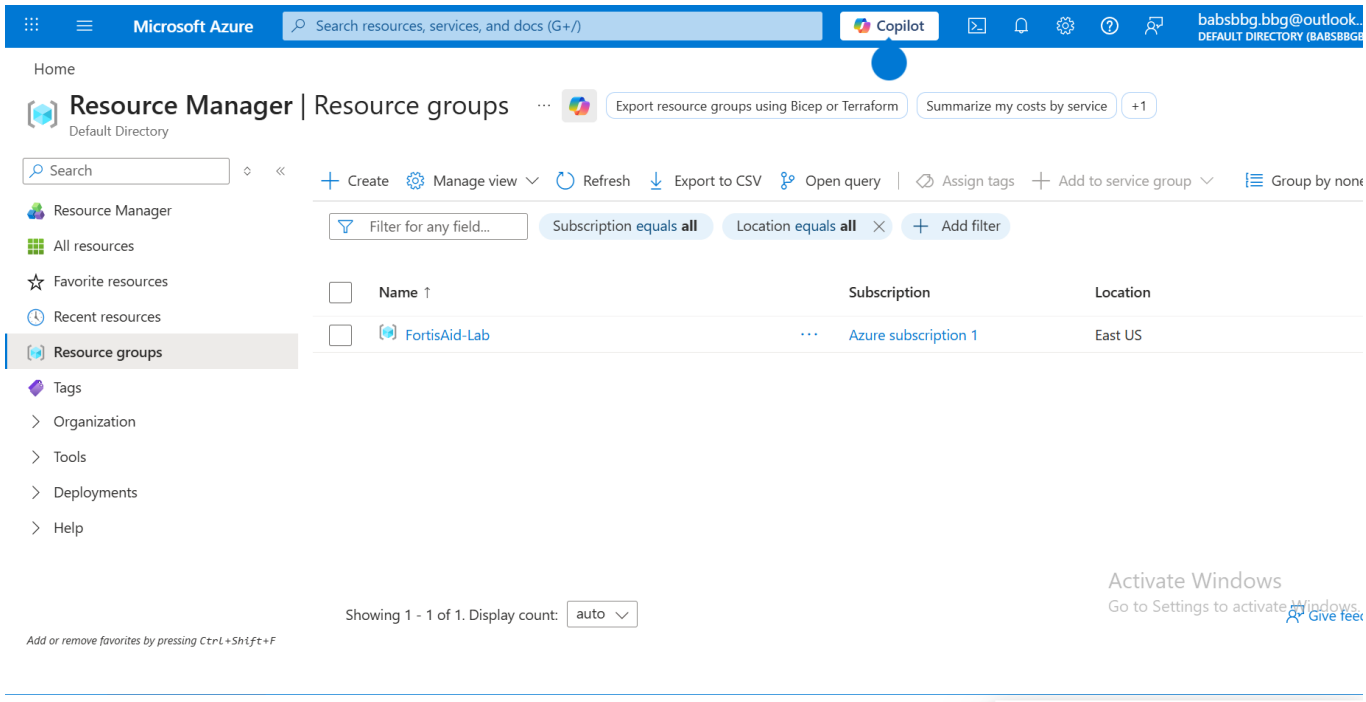
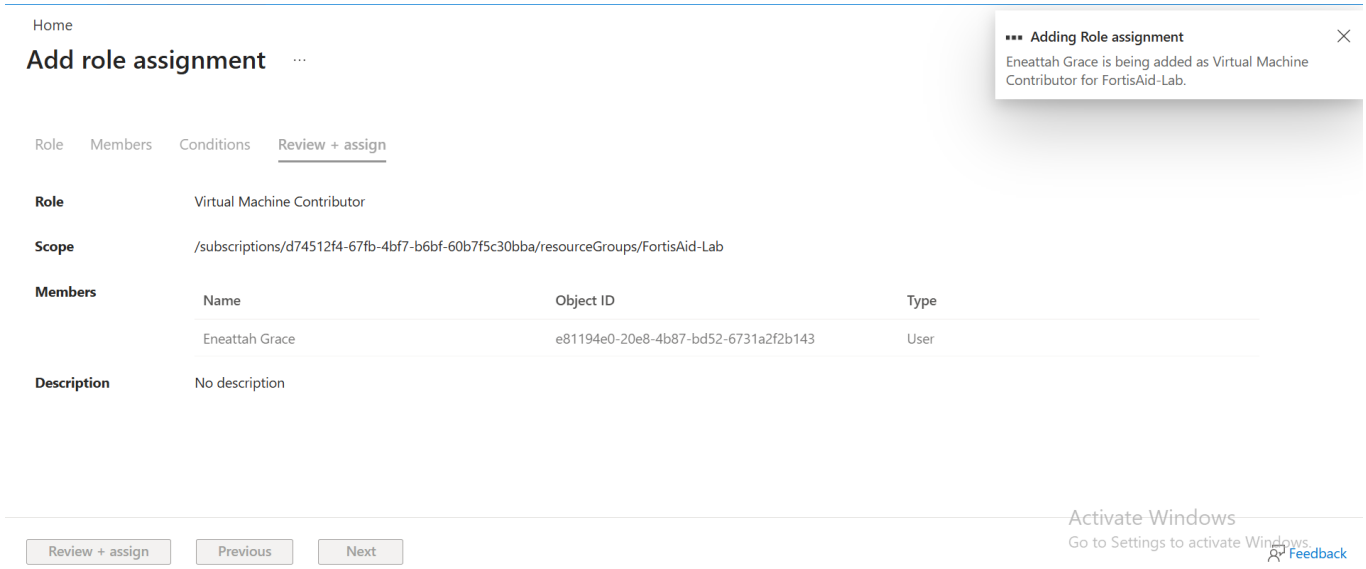
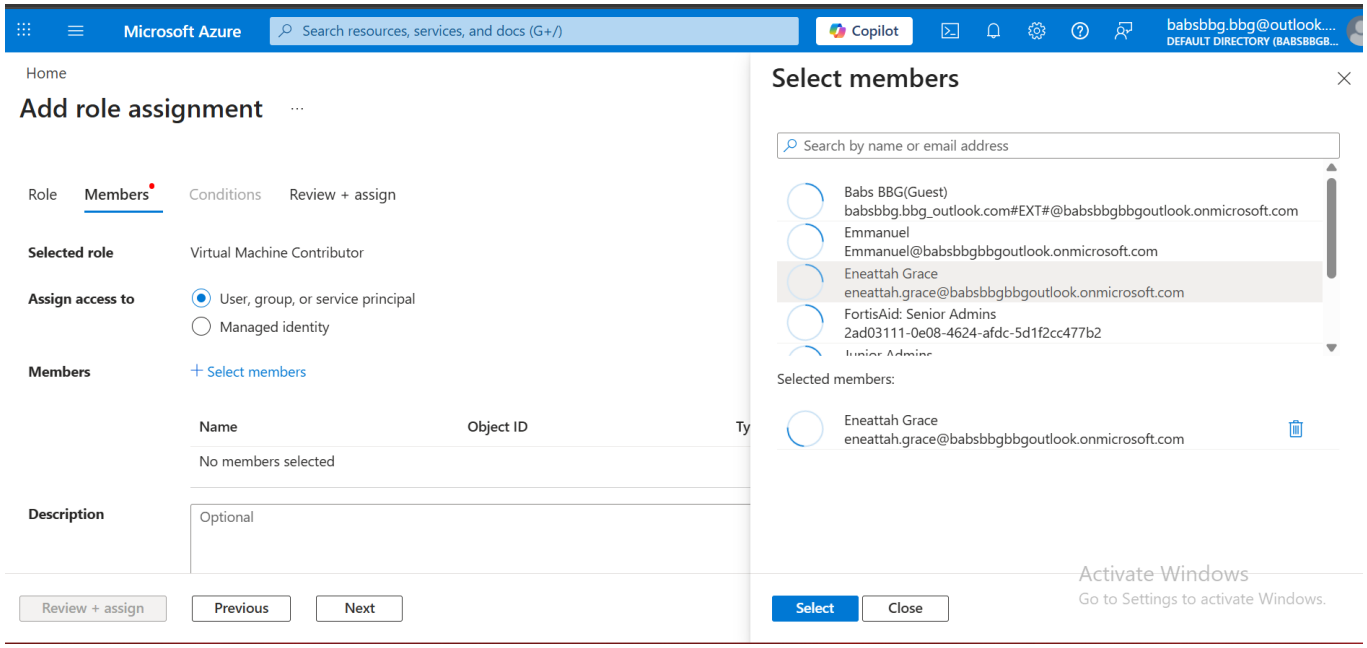




Step 6: Verify effective permissions (Critical)

1. FortisAid-Lab → **Access control (IAM)**
2. **Check access**
3. Search: **Eneattah Grace**
4. Confirm:
   - **Role:** Virtual Machine Contributor
   - **Assignment source:** Service Desk group
   - **Scope:** FortisAid-Lab

This proves the user inherits privileges through group membership.



Why this is security-relevant

- **Least privilege:** Role scoped to a resource group, not the whole subscription
- **Scalable access:** Add/remove members from groups without changing RBAC
- **Audit-friendly:** Group assignments simplify access reviews and compliance checks
- **Zero trust aligned:** No direct user privilege grants, controlled inheritance only

Common Pitfalls Encountered & Resolve

Pitfall 1: Invalid userPrincipalName domain (CLI)

Issue:
User creation failed with:

> "The domain portion of the userPrincipalName property is invalid."

Root cause:
Azure Entra ID only allows user creation using **verified tenant domains**.

Resolution:
Used the tenant's verified domain:

```
babsbbgbbgoutlook.onmicrosoft.com
```

Security takeaway:
Enforcing verified domains prevents identity spoofing and untrusted namespaces.

Pitfall 2: `New-AzADGroup` parameter mismatch (PowerShell)

Issue:
PowerShell returned:

> "A positional parameter cannot be found that accepts argument 'True'."

Root cause:
`New-AzADGroup` does **not** accept `-SecurityEnabled $true`.
Security groups are created by default.

Resolution:
Removed the unsupported parameter.

```
New-AzADGroup -DisplayName "Junior Admins" -MailNickname "JuniorAdmins"
```

Security takeaway:
Understanding cmdlet behavior prevents automation errors and misconfiguration.

Pitfall 3: `az ad domain` command unavailable

Issue:
`az ad domain list` not recognized in Cloud Shell.

Root cause:
Azure CLI has moved some Entra ID functionality behind Microsoft Graph.

Resolution:
Verified domains via:

- Microsoft Entra admin center
- Microsoft Graph (via `az rest`)

Security takeaway:
Identity operations increasingly rely on Graph APIs rather than legacy CLI surfaces.

Pitfall 4: Confusing "az login" behavior in Cloud Shell

Issue:
`az login` prompted device authentication unexpectedly.

Root cause:
Cloud Shell is **already authenticated** using the portal session.

Resolution:
Skipped `az login` and used `az account show` to confirm context.

Security takeaway:
Implicit authentication reduces credential exposure in managed shells.

Final Summary

> This walkthrough demonstrates a secure, enterprise-aligned approach to Azure RBAC using Microsoft Entra ID. By combining portal-based setup with PowerShell and Azure CLI automation, and validating effective access through group inheritance, the project reflects real-world cloud security practices rather than tutorial-driven configuration.